Telmo Cunha

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Profile

MSc graduate in Pure and Applied Mathematics with a background in Engineering Physics and experience in mathematical modelling, machine learning and scientific programming. Seeking junior opportunities in embedded systems, robotics, and autonomous systems, particularly in simulation, software development, and hardware-software integration.

Education

- **University of Lisbon - Instituto Superior Técnico (IST)** — Lisbon, Portugal
  MSc in Applied Mathematics and Computation (Final grade: 17/20) — *2025*
  Thesis topic: Analysis of Nonlinear Wave-Equations (Grade: 19/20)
- **University of Lisbon - Instituto Superior Técnico (IST)** — Lisbon, Portugal
  BSc in Engineering Physics (Final grade: 15/20) — *2019*
  Swiss-European Mobility exchange at École Polytechnique Fédérale de Lausanne (EPFL)

Technical Skills

- **Programming**: Python (proficient), C/C++, R and Assembly (academic experience).
- **Python**: NumPy, SciPy, pandas, scikit-learn, PyTorch, matplotlib.
- **Mathematics/Computing**: Mathematical and numerical modelling, dynamical systems, optimisation, machine learning, statistical/data analysis.
- **Tools**: Mathematica, LaTeX, Git.

Research Experience

- **Research Assistant at Instituto Superior Técnico** — Lisbon, Portugal
  Ergodicity breaking in quantum many-body theory — *03/2026 - 10/2026*
  - Developed Python-based numerical simulations of constrained dynamical systems using Monte Carlo and exact diagonalization methods; analyzed large computational datasets to identify signatures of a phase transition.
- **Summer Research Assistant at Aalto University (AScI Program)** — Helsinki, Finland
  Equivariance properties of Graph Neural Networks (GNNs) — *04/2023 - 09/2023*
  - Developed analytical results concerning equivariance properties of Graph Neural Networks.
- **Summer Research Assistant at Instituto Gulbenkian de Ciência (IGC)** — Oeiras, Portugal
  Evolution of cancer cell populations — *06/2017 - 08/2017*
  - Developed and analysed an ODE mutation-selection model of cancer-cell population dynamics.

Selected Projects

- **Mathematical Foundations of Machine Learning**: Wrote a set of notes covering the foundations of Learning Theory, Machine Learning and Kernel Methods with implementations in Python (repo).
- **Manifold Learning**: Applied manifold learning and nonlinear optimization techniques to dimensionality-reduction problems, with implementations in Python (repo).
- **Feature Selection**: Implemented information-theoretic feature-selection methods in Python and evaluated their effectiveness for dimensionality reduction and predictive modelling (repo).
- **Collaborative Filtering**: Developed a movie recommendation system using matrix-factorisation-based collaborative filtering, implementing model training and evaluation in Python (repo).

Teaching Experience

- **Invited Assistant Professor at ISCTE – Instituto Universitário de Lisboa** — Lisbon, Portugal
  Taught Calculus I, Calculus II and Linear Algebra to first-year Economics students, including classes taught in English. — *2024, 2025*
- **Private Tutor - Mathematics and Physics** — Lisbon, Portugal
  Individual instruction in mathematics and physics to high-school and undergraduate students. — *2015 – 2023*

Additional Training

- **Quantum Landscapes International School**: Paris, France, 2026.
- **Aerospace Information Technology Summer School**: Würzburg, Germany, 2015.

Language Skills

- **Languages**: Portuguese (Native); English (Fluent); Spanish (Basic); French (Basic).